

VRAI Faces — Research Brief RB-003

Avatar Visual Fidelity: Inner-Mouth Geometry + Region Texture Handling for Expression

ID: RB-003 **Created:** 2026-06-05 **Status:** Open — not started **Gates:** Phase 7 (avatar fidelity)
Modules: mesh_builder, shader_translucent, face_ingest
Relates to: RB-001/ADR-0034, ADR-0001, ADR-0005, ADR-0014
Run in: Claude Cowork (deep research)

A follow-up to RB-001. The ARKit-52 deformation rig is accepted and correct (46/52 verified, zero mislabels), but per-shape QA surfaced a class of **texture/topology** artifacts that are inherent to morphing a single static portrait over the 468-vertex mesh. This brief defines the research to close them. Nothing here blocks the current build — the avatar renders, speaks, and emotes; this is fidelity polish.

1. Objective

Eliminate the visual artifacts that appear when the accepted ARKit-52 rig drives the photo-textured 468-vertex mesh — without breaking local-first, deterministic, PHI-safe delivery. Four artifact classes (RB-001 QA, 2026-06-05): (a) the open mouth has **no inner cavity** (teeth/tongue), so jawOpen/mouthClose stretch lip texture into the gap; (b) `tongueOut` has no geometry; (c) **mouth corners tear** on smile/mouthUpperUp/ noseSneer; (d) the **lower eyelid smears** on eye-closure shapes.

2. Why this is gated (not done inline)

- Needs new **geometry** (an inner-mouth cavity with teeth/tongue) and/or **texture** handling beyond the deformation basis — a larger pipeline than RB-001's bake.
- Candidate teeth/tongue assets and any face-texture-synthesis tools carry **licensing** review (some paid/restricted).
- Must preserve local-first (ADR-0001), deterministic/auditable output (ADR-0005), and the PHI posture (ADR-0014 — never gather/scrape faces; portraits stay consented + local).

3. Current state (context for the researcher)

- mesh_builder** builds the real 468-vertex MediaPipe mesh; **morph_basis** applies the baked ARKit-52 (+ `eyesClosed`) deltas; the rig is accepted (RB-001 QA).
- The mesh is textured with the **single consented portrait** (UVs = the detected landmark x,y). There is **no inner-mouth geometry** and no separate eye/lip texture, so deforming

stretches/tears/smears the one static image at high-motion regions.

- **shader_translucent** is a `MeshPhysicalNodeMaterial` (TSL) under WebGPU.

4. Research questions

1. **Inner mouth:** how to add a teeth/tongue/oral-cavity that tracks `jawOpen` — a secondary proxy mesh, a baked cavity in the topology, or a shader trick — that is license-clean, deterministic, and small to bundle?
2. **Texture at deformation extremes:** how to stop the corner-tear and eyelid-smear — per-region textures, UV relaxation, an inpainted/synthesized skin texture, or a hybrid photo + procedural-skin shader? Trade-offs vs the photoreal look.
3. **tongueOut:** synthesize a deterministic tongue protrusion (geometry + the AU) absent from ICT-FaceKit.
4. **Mouth-corner topology/UV:** is the tear a UV seam, a triangle inversion, or pure texture stretch — and what's the minimal fix?
5. Which of these are achievable on-device at 60 fps under WebGPU + the translucent shader?

5. Evaluation criteria

Criterion	Weight	Notes
License — clinical + commercial + redistributable	MUST-PASS	Any teeth/tongue asset or texture tool must be bundleable in a shipped on-device app.
Determinism & PHI safety	MUST-PASS	Deterministic (ADR-0005); no facial-image gathering/scraping; nothing leaves device (ADR-0014).
Artifact reduction (the 4 classes)	High	Inner-mouth + corner-tear + eyelid-smear + tongueOut, ranked by clinical visibility (inner-mouth shows in every open-mouth viseme — highest).
Runtime cost (WebGPU, 60 fps)	High	Must hold the \$5 frame budget on the target tablets (ADR-0032 hardware).
Bundle size / local-first	Medium	Assets served from app origin (ADR-0001), never a runtime CDN.
Effort / maintenance	Medium	Prefer a one-time offline bake + a drop-in, like RB-001.

6. Deliverables (what Cowork should return)

- A ranked recommendation per artifact class (inner-mouth, corner-tear, eyelid-smear, tongueOut).

- A concrete integration path (assets + module touches: `mesh_builder` / `shader_translucent` / a new oral-cavity piece), with bundle-size + runtime-cost estimates.
- A license summary + any procurement; a clear GO / NO-GO with rough effort.

7. How the result re-enters the build

Research output → a new ADR (next number in `Memory_management.MD §7`) → implementation: the inner-mouth geometry / texture handling drops into `mesh_builder` + `shader_translucent`; the rig (RB-001) and the drive chain (speech→viseme/emotion → `animation_runtime`) are unchanged.

8. Hard constraints

- **Local-first (ADR-0001):** assets served from the app origin; no runtime CDN.
- **Deterministic & auditable (ADR-0005):** reproducible geometry/textures.
- **PHI (ADR-0014):** nothing leaves the device; never gather or scrape facial images; portraits remain facilitator-supplied + consented.
- **Naming/contract:** keep the ARKit-52 + `eyesClosed` morph names; no contract change to the rig.

VRAI Faces (MedSim V8) · Research briefs live in `vrai-faces/research/` and follow the research-driven-enhancement strategy in `docs/ROADMAP.md`. Render the PDF with `node packages/core/scripts/render-pdf.mjs` `research/RB-003_avatar-visual-fidelity.html` `research/RB-003_avatar-visual-fidelity.pdf`.